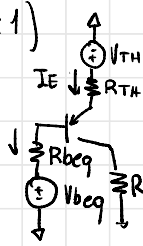


Ex 1) equivalent Thevenin scheme

IV CALL 2024



$$R_{TH} = R_F // R_E = 903 \Omega$$

$$V_{TH} = V_{DD} \cdot \frac{R_F}{R_F + R_E} + V_O \cdot \frac{R_E}{R_F + R_E} = 9,545 V$$

$$\begin{cases} V_{BEQ} = 6V \\ R_{BEQ} = 60 k\Omega \end{cases}$$

$$-V_{BEQ} - R_{BEQ}I_B + V_{BE1} - R_{TH}(B\beta + 1)I_B + V_{TH}$$

$$I_B = - \frac{V_{BEQ} - V_{BE1} - V_{TH}}{R_{BEQ} + R_{TH}(B\beta + 1)} = 14.42 \mu A$$

$$I_C = 2,163 mA$$

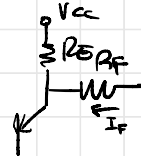
CALCULATION OF I_D

$$-R_C \cdot I_C + V_{GS2} + R_S I_{DSS} \left(\frac{V_{GS}}{V_T} - 1 \right)^2 = 0$$

$$V_{GS}^2 \left(\frac{R_S I_{DSS}}{V_T^2} \right) + V_{GS2} \left(1 - \frac{2 R_S I_{DSS}}{V_T} \right) + R_S I_{DSS} - R_C I_C = 0$$

$$V_{GS} \begin{cases} 5,436 V \checkmark \\ -14,42 \times \end{cases}$$

$$I_D = 593.4 \mu A$$



CALCULATION OF R_D

$$V_O = R_F I_F - V_{BE1} + R_{BEQ} I_B + V_{BEQ}$$

$$I_F = \frac{V_O + V_{BE1} - V_{BEQ} - R_{BEQ} I_B}{R_F} = -256.52 \mu A$$

$$R_D = \frac{V_{DD} - V_O}{I_D - |I_F|} = 14842 \Omega$$

$$-V_{TH} + R_{TH} I_E - V_{CE} + R_D I_D = 0$$

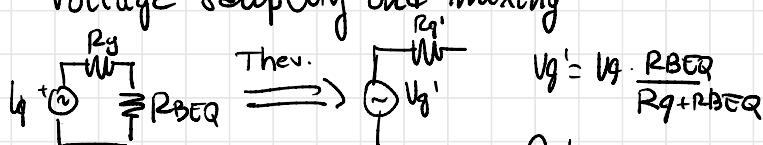
$$V_{DS} = V_O - R_S I_D = 4,596 V$$

$$V_{CE} = -V_{TH} + R_{TH} I_E + R_D I_D = -1,725 V$$

$$g_{m1} = 83.19 \text{ mS} \quad g_{m2} = 487.2 \text{ pS}$$

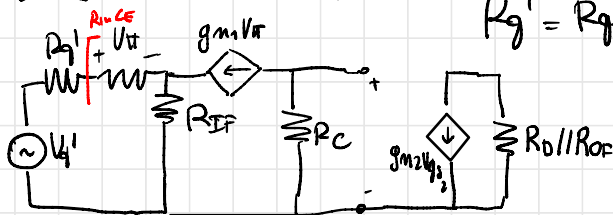
$$R_{\pi1} = 1803 \Omega$$

Voltage sampling and mixing



$$R_{g'} = R_g // R_{BEQ} = 983.606 \Omega$$

$$\begin{cases} \beta\text{-network} \\ R_F \\ R_E \\ R_{IF} = R_E // R_F = 903 \Omega \\ R_{OF} = R_E + R_F = 11 \text{ k}\Omega \\ \beta = \frac{R_E}{R_E + R_F} = 0.091 \end{cases}$$



OPEN LOOP GAIN ($\frac{V_o}{V_{g'}}$)

$$\frac{V_o}{V_{gs2}} = -g_{m2}(R_D // R_{OF}) = -3.078$$

$$\frac{V_{gs2}}{V_{t1}} = -g_{m1}R_C = -224.61$$

$$V_{t1} = R_{\pi1} \cdot \frac{V_{G1}}{R_{g'} + R_{INCE}} \Rightarrow \frac{V_{t1}}{V_{G1}} = \frac{R_{\pi1}}{R_{g'} + R_{INCE}} = 0.012875 \dots$$

$$R_{INCE} = R_{\pi1} \cdot (1 + R_{IF}(\frac{1}{R_{\pi1}} + g_{m1})) = 133054 \Omega$$

$$\beta A_{VOL} = 0.810$$

$$A_{VOL} = \frac{V_o}{V_{gs2}} \cdot \frac{V_{gs2}}{V_{t1}} \cdot \frac{V_{t1}}{V_{G1}} = 8.901 \quad A_V^{CL} = \frac{A_V^{OL}}{1 + \beta A_{VOL}} = 4.917$$

$$A_V(\frac{V_o}{V_{g'}}) = A_V^{CL} \cdot \frac{R_{BEQ}}{R_{BEQ} + R_{g'}} = 4.837$$

$$R_{in}^{OL} = R_{g'} + R_{INCE} = 140037.6 \Omega$$

$$R_{in}^F = R_{in}^{OL} (1 + \beta A_{OL}) = 253468 \Omega$$

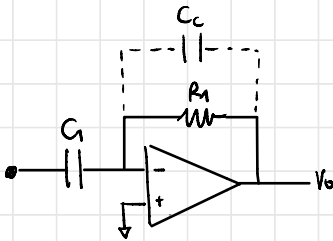
$$R_{in} = (R_{in}^F - R_{g'}) // R_{BEQ} = 48479.4 \Omega$$

5) f_L due to C_g applying SCTC

$$R_{S^g} = R_g + R_{in}$$

$$f_L = \frac{1}{2\pi \cdot R_{S^g} \cdot C_g} = 32.16 \text{ Hz} \rightarrow \text{ALREADY CONSIDERED IN CLOSED LOOP?}$$

Ex 2)



$$Z_2 = R_1$$

$$Z_1 = \frac{1}{sC_g}$$

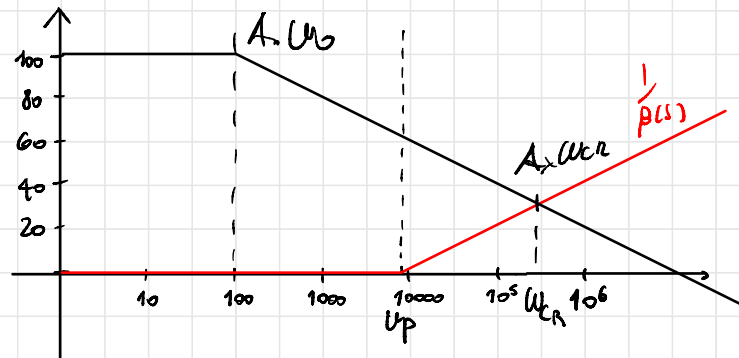
Ideal gain $A_v(s) = -\frac{Z_2}{Z_1} = -\frac{R_1}{\frac{1}{sC_g}} = -sC_g R_1$

NON INVERTING

β_{net} (With C_c OPEN)

$$\beta(s) = \frac{\frac{1}{sC_g}}{\frac{1}{sC_g} + R_1} = \frac{1}{1 + sC_g R_1}$$

$$\omega_p = \frac{1}{C_g R_1} = 8333 \text{ rad/s}$$



$$\begin{cases} A_0 \omega_0 = A_x \omega_{CR} \\ \frac{1}{\beta(0)} \cdot \frac{1}{\omega_p} = A_x \cdot \frac{1}{\omega_{CR}} \end{cases}$$

$$\omega_{CR} = \sqrt{A_0 \omega_0 \omega_p \beta(0)}$$

$$= 288669.3 \text{ rad/s}$$

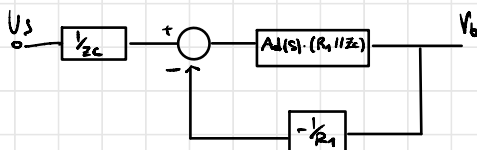
$$(45943.15 \text{ Hz})$$

$$PM^0 = 180 - \tan^{-1}\left(\frac{\omega_{cr}}{\omega_p}\right) - \tan^{-1}\left(\frac{\omega_{cr}}{\omega_1}\right) = 1.67^\circ \rightarrow \text{Very low PM!}$$

Block diagram

Current mixing - Voltage sampling

$$Z_c = \frac{1}{sC}$$



β -network
 $\frac{R_1}{R_1 + Z_c}$

New β -network with C_c inserted

$$\beta(s) = \frac{\frac{1}{sC_c}}{\frac{1}{sC_1} + \frac{R_1}{1+sC_cR_1}} = \frac{1+sC_cR_1}{1+sR_1(C_c+C_1)}$$

$$\omega_z = \frac{1}{C_c R_1} \quad \omega_p' = \frac{1}{R_1(C_c+C_1)}$$

To have 45° of PM $\rightarrow \omega_z = \omega_{cr}$

$$C_c = \frac{1}{\omega_{cr} \cdot R_1} = 288.6 \text{ pF}$$

$$\omega_p' = 8100 \text{ rad/s} \quad PM' = 180 - \tan^{-1}\left(\frac{\omega_{cr}}{\omega_o}\right) - \tan^{-1}\left(\frac{\omega_{cr}}{\omega_p'}\right) + \tan^{-1}\left(\frac{\omega_{cr}}{\omega_z}\right) = 46.24^\circ$$

$$\omega_{cr}^C = \sqrt{A_o U_o \cdot \omega_p'} = 284605 \text{ rad/s}$$

